11.2 Areas of Circles and Sectors

Name _____ Period ____ Date _____

Geometry 11.2 · Student sheet

I can…

- find the area of a circle and use it to find a radius.
- find the area of a sector of a circle.
- find the area of a shaded region made of circles and polygons.

Vocabulary · fill in the missing word

Sector of a circle — the region bounded by two radii of the circle and their _____ arc

Area of a circle — the number of _____ units inside the circle: $A = \pi r^2$

Part 1 · Discovery: a circle becomes a parallelogram

Cut your circle into 16 equal sectors and arrange them, alternating up and down, into a shape like a parallelogram.

Radius of my circle	Base of the "parallelogram"	Height	Base × height	π × radius²

The base is half the _____, and the height is the _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Area of a circle: A = _____

Example 1

Find the area of each circle. Round to the nearest hundredth.

a. radius 2.5 centimeters

b. diameter 10 inches

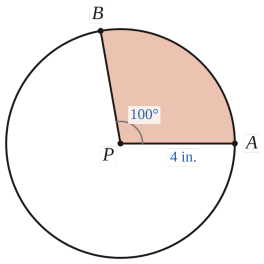
Example 2

A circular rug has an area of 113.1 square feet. Find its radius and diameter. Round to the nearest tenth.

Area of a sector: area of sector APB = (m(arc AB) ÷ 360°) · _____

Example 3

Find the area of the shaded sector and of the unshaded sector. Round to the nearest hundredth.



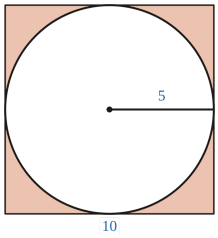
Example 4

A sector with a central angle of 80° has an area of 35 square meters. Find the area of the whole circle and its radius. Round to the nearest hundredth.

Shaded regions: area of the shaded region = area of the whole figure _____ the area of the part that is not shaded.

Example 5

A circle is inscribed in a square with sides of 10. Find the area of the shaded region. Round to the nearest hundredth.



Part 3 · Partner practice

Partner 1

A circle has a radius of 9 feet. Find the area of a sector with a central angle of 40°. Round to the nearest hundredth.

Partner 2

A circle with a diameter of 8 is inscribed in a square with sides of 8. Find the area of the region inside the square but outside the circle. Round to the nearest hundredth.